

PROBING TRANSFORMER REASONING TRANSFER THROUGH ALPHABET EQUIVALENCE

Anonymous authors

Paper under double-blind review

ABSTRACT

A reasoner is *abstract* if it applies a general procedure to a problem’s structure, indifferent to how that structure is concretely instantiated; it is *concrete* if what it learned is tied to the particular instantiation trained on. This gives a testable marker: an abstract reasoner should transfer mastery of one problem instance to another. We study equivalence via alphabet renaming, and test transfer using a weight-transplant control – copying trained per-symbol weights onto the new instance without retraining – distinguishing genuine reasoning failure from symbols never trained on. We instantiate this with Sudoku, where an equivalent puzzle is any relabeling of its digits. A transformer trained on one such puzzle transfers to an equivalent one no better than an untrained network – yet the transplant control recovers its original performance exactly. Reasoning, in that narrow sense, is abstract; recognizing that a new instance calls for it is not. We probe this recognition gap three ways. Freezing the shared network and adapting only a new alphabet’s parameters fares worse than training an unfrozen model from scratch on the same data: prior exposure to other alphabets does not ease learning a new one. Recombining already-known symbols into unfamiliar combinations yields graded, partial success – above chance, short of the seamless transfer an abstract reasoner should show, degrading as combinations diverge from training. Replacing the model’s fixed, per-alphabet output vocabulary with a mechanism that selects among a puzzle’s own visible symbols helps substantially with recombination, but only marginally with transfer to a truly unseen alphabet. These results isolate a specific failure to recognize alphabet-equivalent instances as such, distinct from a general failure to reason, and show several natural interventions narrow, but do not close, this gap.

1 INTRODUCTION

A model that has learned to solve a problem can be said to reason *abstractly* about it if the procedure it applies operates on the problem’s structure, indifferent to how that structure happens to be dressed up in a particular presentation. It reasons only *concretely* if what it has learned is tied to the specific surface form it was trained on, rather than to the structure underneath. This distinction is central to any claim that a learned system reasons rather than merely fits patterns in its training distribution, but such claims are frequently made without a test that could actually distinguish the two possibilities. Benchmarks such as SCAN (Lake & Baroni, 2018) and COGS (Kim & Linzen, 2020) test a closely related property – generalizing to unseen *combinations* of familiar components – and have repeatedly found large gaps between in-distribution and compositional-generalization accuracy in standard neural architectures. We study a different, narrower notion of generalization: not new combinations of familiar parts, but the exact same combination presented under a purely relabeled vocabulary, which isolates a more basic question – does the model recognize the relabeled instance as the problem it already knows how to solve at all.

We propose a concrete, checkable operationalization of this distinction. An abstract reasoner should treat two different instances of the same underlying problem interchangeably: having mastered one, it should transfer to solving the other, since its procedure does not depend on which particular instantiation it happens to face. We instantiate “different instances of the same problem” through the simplest transform that leaves a problem’s structure completely intact while changing its surface

form entirely: renaming its symbol alphabet. Two problems P and Q are *equivalent* under this definition when Q is obtained from P purely by relabeling which symbols represent which underlying roles – nothing about the structure, constraints, or solution changes, only the names used to present it.

This gives more than a single empirical test: it gives a general recipe applicable to any symbolic or combinatorial domain with an analogous notion of relabeling. The recipe has two parts. First, define equivalence via a checkable relabeling, and measure whether a model trained on one instance transfers, unaided, to an equivalent one. Second – and this is the part that is easy to get wrong – use a targeted control to separate two failure modes that a naive transfer evaluation cannot distinguish on its own. A model may fail on a new instance either because its underlying reasoning does not generalize, or simply because the new instance’s symbols were never seen during training, so any parameters associated with them are untrained noise regardless of how good the model’s reasoning otherwise is. We isolate the second explanation with a *weight-transplant* control: after training, we copy the parameters associated with each of P ’s symbols directly onto the parameters for Q ’s corresponding symbols, using the known-by-construction correspondence, with no further training. This tells us how the model would perform on Q if it already knew the correspondence between the two instances’ symbols – precisely the information a naive zero-shot evaluation cannot provide on its own.

We instantiate this recipe concretely using Sudoku: a well-defined, verifiable, and cheaply generated combinatorial puzzle whose solution is invariant to any consistent renaming of its 9 symbols, making it a clean testbed for this question. Training a transformer to solve Sudoku puzzles written in one symbol alphabet, we find that it transfers to an equivalent puzzle in a different alphabet at chance level – performance indistinguishable from a network that was never trained at all – even though the weight-transplant control recovers its original accuracy *exactly*. This is the paper’s central result: the model’s procedure does reuse across equivalent instances once it is told how their symbols correspond; what it lacks is any mechanism for discovering that correspondence on its own. We then probe this recognition gap along three further axes – pretraining diversity, an architectural change that removes a specific structural bottleneck, and training-time data diversity at a much larger scale – and show that each narrows, but does not close, the gap, with a further finding that training diversity resolves the gap into two distinct sub-problems with different answers: recombining already-known symbols is solved by sufficient diversity, while transfer to a genuinely novel symbol is not.

Contributions.

- An operationalized, reusable methodology for testing knowledge transfer between equivalent instances of a symbolic problem, including a weight-transplant control that isolates genuine reasoning-transfer failure from the confound of never-trained parameters.
- Applied to Sudoku, we show naive transfer across alphabet-equivalent puzzles fails completely – statistically indistinguishable from an untrained network – while the transplant control recovers full performance exactly, demonstrating the failure is one of recognition, not of reusable computation (Section 4.1).
- We show this recognition gap persists across two further natural interventions: pretraining diversity across several known alphabets, and a redesigned architecture that removes a specific structural bottleneck that prevents zero-shot generalization by construction – each narrows but does not close the gap (Section 4.2–3).
- We show training-time data diversity resolves this gap into two distinct sub-problems with different answers: recombining already-known symbols in novel combinations is essentially solved by sufficient training diversity, while transfer to a genuinely novel, never-seen symbol remains unsolved by every intervention tested (Section 4.4).

2 RELATED WORK

Neural Sudoku solving. Palm et al. (2018) introduced Recurrent Relational Networks, which solve Sudoku via iterative message passing between cell-level representations on a graph – the architectural lineage our encoder-based model, and its optional iterative-refinement mechanism, belong to. That work, and subsequent neural combinatorial solvers, focus on solving accuracy within

a single fixed vocabulary; we instead hold solving accuracy on the training vocabulary fixed near ceiling throughout, and ask what happens under a symbol relabeling that leaves the underlying combinatorial problem completely unchanged.

Compositional and systematic generalization. SCAN (Lake & Baroni, 2018) and COGS (Kim & Linzen, 2020) test whether sequence models generalize to novel *combinations* of familiar primitives, and both find persistent, large gaps between in-distribution and compositional accuracy across a range of architectures. Our question is narrower and, in a specific sense, easier: we do not ask the model to combine familiar parts in new ways, only to recognize the exact same combination it was trained on, wearing a different vocabulary. That we still find a total failure to recognize this – indistinguishable from an untrained network – suggests the difficulty compositional-generalization benchmarks document may in part reflect an even more basic failure to abstract away from surface symbol identity at all.

Symbol binding and variable binding. Webb et al. (2021) is the closest direct precedent for our framing: their Emergent Symbol Binding Network uses an indirection mechanism over external memory that lets learned rules generalize near-perfectly to entirely novel filler entities, given only a handful of examples, because its computation operates on relations between representations rather than their identity. Altabaa et al. (2024) formalize a version of this idea for Transformers directly, using a relational cross-attention mechanism architecturally constrained to depend only on similarity between representations, never their content. Our Section 4.3 pointer-based redesign is a much simpler, less architecturally constrained step in the same direction – removing a fixed, identity-keyed output vocabulary rather than building in an explicit relational bottleneck – and our results (helping substantially with one sub-problem, only marginally with the other) suggest the stronger, more explicit mechanism of Altabaa et al. (2024) may be necessary where our simpler intervention is not sufficient (see Section 4.4 and the Discussion).

Pointer and copy mechanisms. Our Section 4.3 architecture, which predicts a blank cell’s value by selecting among the symbols visible elsewhere in the same puzzle instance rather than classifying into a fixed global vocabulary, is a direct application of Pointer Networks (Vinyals et al., 2015) and pointer-generator copy mechanisms (See et al., 2017) to this setting. Both were originally motivated by output spaces that vary with the input (sorting an input sequence; naming a rare word from the source document) rather than by the alphabet-equivalence question we study; we use the same mechanism for a different purpose, to remove the specific structural obstacle – a per-alphabet output vocabulary – that makes zero-shot transfer to a new alphabet impossible by construction in the naive classifier design.

Symbolic mechanisms in pretrained language models. Contemporaneously with this direction of work, Yang et al. (2025) identify an emergent symbolic architecture inside pretrained large language models – symbol abstraction, symbolic induction, and retrieval heads operating on abstract variables rather than raw tokens – that supports abstract reasoning in a related sense. Our study is complementary: rather than probing an existing large pretrained model, we train small transformers from scratch under fully controlled data, isolating the specific alphabet-relabeling manipulation from every other confound a pretrained model’s training history would introduce.

Positioning. To our knowledge, the specific combination we use – defining instance equivalence via a checkable symbol relabeling, and a weight-transplant control that isolates genuine reasoning-transfer failure from the confound of never-trained parameters – has not been used together in this form, though each piece draws on the established ideas above.

3 PROBLEM SETUP AND METHOD

4 EXPERIMENTS AND RESULTS

5 DISCUSSION

6 LIMITATIONS

7 CONCLUSION

REFERENCES

- Awni Altabaa, Taylor Webb, Jonathan Cohen, and John Lafferty. Abstractors and relational cross-attention: An inductive bias for explicit relational reasoning in Transformers. In *International Conference on Learning Representations*, 2024.
- Najoung Kim and Tal Linzen. COGS: A compositional generalization challenge based on semantic interpretation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 9087–9105, 2020.
- Brenden Lake and Marco Baroni. Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. In *International Conference on Machine Learning*, pp. 4487–4499, 2018.
- Rasmus Berg Palm, Ulrich Paquet, and Ole Winther. Recurrent relational networks. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Abigail See, Peter J. Liu, and Christopher D. Manning. Get to the point: Summarization with pointer-generator networks. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1073–1083, 2017.
- Oriol Vinyals, Meire Fortunato, and Navdeep Jaitly. Pointer networks. In *Advances in Neural Information Processing Systems*, volume 28, pp. 2692–2700, 2015.
- Taylor W. Webb, Ishan Sinha, and Jonathan D. Cohen. Emergent symbols through binding in external memory. In *International Conference on Learning Representations*, 2021.
- Yukang Yang, Declan Campbell, Kaixuan Huang, Mengdi Wang, Jonathan Cohen, and Taylor Webb. Emergent symbolic mechanisms support abstract reasoning in large language models. In *International Conference on Machine Learning*, 2025.